

Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 4

Clinical and Research devices often require wired connections, but wireless is taking over as security is

addressed.

ActiGraph.

Hardware Product Evolution



Sometimes the data are stored in a mixed format, text and low level binary

Log Record Types

Note that some undocumented records are used for internal state or testing. They may be safely ignored.

ID (dec)	ID (hex)	Type	Description
0	0x00	ACTIVITY	One second of raw activity samples packed into 12-bit values in XYZ order.
2	0x02	BATTERY	Battery voltage in millivolts as a little-endian unsigned short (2 bytes).
3	0x03	EVENT	Logging records used for internal debugging. These records notably contain information about idle sleep mode. When entering idle sleep mode, a record with payload 0x08 is created. When exiting idle sleep mode, a record with payload 0x09 is created.
4	0x04	HEART_RATE_BPM	Heart rate average beats per minute (BPM) as one byte unsigned integer.
5	0x05	LUX	Lux value as a little-endian unsigned short (2 bytes).
6	0x06	METADATA	Arbitrary metadata content. The first record in every log is contains subject data in JSON format.
7	0x07	TAG	13 Byte Serial, 1 Byte Tx Power, 1 Byte (signed) RSSI
9	0x09	EPOCH	60-second epoch data
11	0x0B	HEART_RATE_ANT	Heart Rate RR information from ANT+ sensor.
12	0x0C	EPOCH2	60-second epoch data
13	0x0D	CAPSENSE	Capacitive sense data
14	0x0E	HEART_RATE_BLE	Bluetooth heart rate information (BPM and RR). This is a Bluetooth standard format.
15	0x0F	EPOCH3	60-second epoch data

Typically time based

- ▶ A header often defines parameters
- ▶ Each row is an entry
- ▶ Each column is a measure

Typical Analysis

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Clinical Standards
of Wearable
Devices (PA)

Devices (Actigraphy)

Data

Typical Analysis

Analysis

Central Tendency

Measures of Variance

Measures of Variance

Measures of Variance

Standardized Measures

Comparison Tests

Analysis of Variance

Repeated Measures ANOVA

- ▶ Measures of central tendency
- ▶ Measures of variability
- ▶ Comparisons
- ▶ Trends (Binned, Change Point, Detrended Fluctuation)

Central Tendency

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- ▶ Arithmetic Mean:

$$\bar{A} = \frac{1}{n} \sum_{i=1}^n a_i$$

- ▶ Geometric Mean¹:

$$\bar{G} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} = \sqrt[n]{x_1 x_2 \dots x_n}$$

- ▶ Harmonic Mean²:

$$\bar{H} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

- ▶ Median

- ▶ Mode

¹useful whenever the quantities to be averaged combine multiplicatively

²most appropriate average for ratios and rates such as speeds, robust to outliers

Measures of Variance

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Clinical Standards of Wearable Devices (PA)

Analysis

Measures of Variance

- Range:

$$\text{range} = \max(X) - \min(X)$$

- Variance:

$$\text{Var}(X) = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

- Standard Deviation:

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$

- ▶ Pooled Standard Deviation:

$$\sigma_p = \sqrt{\frac{\sum_{i=1}^k (n_i - 1) \sigma_i^2}{\sum_{i=1}^k (n_i - 1)}} = \sqrt{\frac{(n_1 - 1) \sigma_1^2 + (n_2 - 1) \sigma_2^2 + \dots + (n_k - 1) \sigma_k^2}{n_1 + n_2 + \dots + n_k - k}}$$

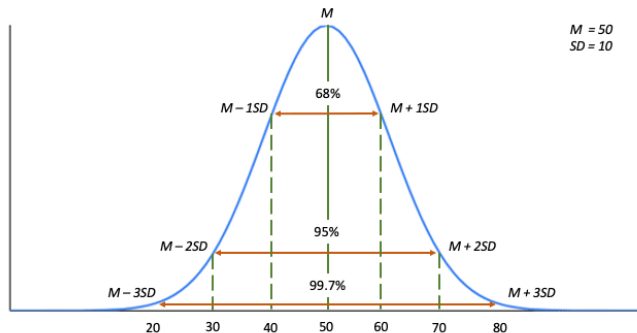
Standard Deviation

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This is a “normal” curve, coming from theory driven by observations. One standard deviation is the inflection point on the curve.

Standard deviations in a normal distribution



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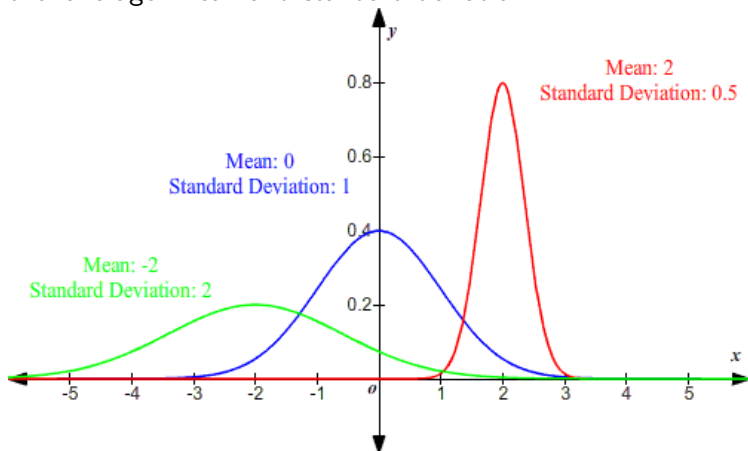
Comparison Tests

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Repeated Measures ANOVA

Pooled Standard Deviation

Suppose we have three participants and have measured the number of steps they take every day for a month. We have an average for each, and a standard deviation around those means for each participant. So then, how do we calculate the 'average' mean and standard deviation?



Standardized Measures (Z-Score)

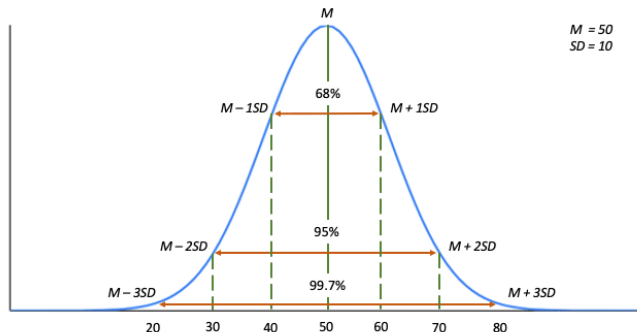
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What if we wanted to make some comparisons when measures are in different units? Or what if the samples seem to have some bias, and we want to understand deviations from that bias?

$$Z_i = \frac{x_i - \mu}{\sigma}$$

Standard deviations in a normal distribution



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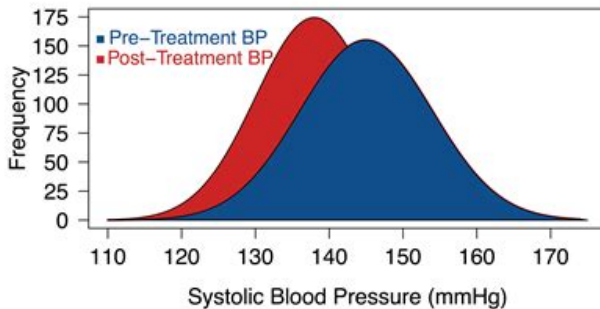
Analysis of Variance

Repeated Measures ANOVA

Suppose that we have samples of blood pressure before a treatment, and then after.

Likewise, what if we had two different groups we wanted to compare?

Systolic Blood Pressure Before and After Treatment



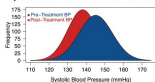
Comparison Tests

Suppose that we have samples of blood pressure before a treatment, and then after.

Likewise, what if we had two different groups we wanted to compare?

We can use a t-test! This tests if the two samples are likely to have been sampled from the same population.

Systolic Blood Pressure Before and After Treatment



$$df = n_1 + n_2 - 2$$

$$\sigma_p = \sqrt{\frac{(n_1-1)\sigma_1^2 + (n_2-1)\sigma_2^2}{df}}$$

$$t = \frac{(\mu_1 - \mu_2)}{\sigma_p \cdot \sqrt{(\frac{1}{n_1}) + (\frac{1}{n_2})}}$$

$$cdf = tcdf(t, df)$$

df	Upper tail areas				
	0.1	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	30.821	63.687
2	1.886	2.920	4.303	6.965	9.525
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.385	1.833	2.262	2.821	3.250
10	1.377	1.812	2.228	2.764	3.183
11	1.369	1.796	2.201	2.718	3.136
12	1.366	1.782	2.179	2.681	3.099
13	1.360	1.771	2.160	2.650	3.067
14	1.356	1.761	2.145	2.624	3.037
15	1.354	1.753	2.133	2.602	3.010
16	1.351	1.746	2.123	2.583	2.991
17	1.349	1.740	2.114	2.567	2.976
18	1.348	1.734	2.106	2.552	2.963
19	1.346	1.729	2.100	2.539	2.951
20	1.345	1.725	2.096	2.528	2.941
21	1.343	1.721	2.090	2.518	2.931
22	1.342	1.717	2.085	2.509	2.923
23	1.341	1.714	2.080	2.500	2.915
24	1.340	1.711	2.076	2.492	2.907
25	1.339	1.708	2.073	2.485	2.900
26	1.338	1.706	2.069	2.479	2.894
27	1.338	1.703	2.065	2.473	2.889
28	1.337	1.701	2.062	2.467	2.884
29	1.336	1.699	2.060	2.462	2.880
30	1.336	1.697	2.057	2.457	2.876
40	1.303	1.664	2.023	2.423	2.794
45	1.296	1.657	2.000	2.390	2.660
120	1.289	1.650	1.980	2.350	2.617
∞	1.282	1.645	1.960	2.326	2.576

Adapted from Table B7 of R. A. Fisher and F. Yates, *Statistical Tables for Biological, Agricultural and Medical Research*, Sixth Edition, Pearson Education Limited, © 1963 R. A. Fisher and F. Yates.

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Analysis of Variance (ANOVA)

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Analysis of Variance

Repeated Measures ANOVA

And what if we have three or more groups? A t-test isn't really going to work for us... why?

We can test if there exists a statistically significant difference among the groups first.

group	1	2	3
	15	17	9
	12	5	12
	12	15	11
	9	12	11
		20	8
		14	
n	5	7	6
Σ	49	85	54

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_1 : H_0 \text{ is untrue}$$

$\alpha = 0.05$ (1/20 probability
what we observe is from
chance alone)

And what if we have three or more groups? A t-test isn't really going to work for us... why?

We can test if there exists a statistically significant difference among the groups first.

group	1	2	3
	15	17	9
	12	5	12
	12	15	11
	9	12	11
		20	8
		14	
n	4	6	5
$\sum$	48	83	51

$$N = \sum_{i=1}^k n_i = 15$$

Total Sum of Squares:

$$SS_T = \sum (X - \bar{X})^2$$
$$= \sum X^2 - \frac{(\sum X)^2}{N}$$

Sum of Squares Between Groups:

$$\begin{aligned} SS_B &= \sum n_j (\bar{X}_j - \bar{X})^2 = \\ &= \sum_{i=1}^k \left(\frac{(\sum X_i)^2}{n_i} \right) - \frac{(\sum X)^2}{N} \\ &= \frac{48^2}{4} + \frac{83^2}{6} + \frac{51^2}{5} - \frac{182^2}{15} \end{aligned}$$

Sum of Squares Within Groups:

$$SS_W = SS_T - SS_B$$

And what if we have three or more groups? A t-test isn't really going to work for us... why?

We can test if there exists a statistically significant difference among the groups first.

MS is our mean squares

source of variation	SS	df	MS	F
between groups	SS_B	$k - 1 = 2$	$\frac{SS_B}{2}$	$\frac{MS_B}{MS_W}$
within groups (error)	SS_W	$N - k =$	$\frac{SS_W}{12}$	

And then use an F – *stat* lookup table / function.

Clinical Standards of Wearable Devices (PA)

Analysis

Analysis of Variance

Suppose we have made measures at several time points on a group of subjects, such that we have something like this:

source of variation	SS	df	MS	F
between groups	SS_B	$k - 1 = 2$	$\frac{SS_B}{2}$	$\frac{MS_B}{MS_W}$
within groups (error)	SS_W	$N - k =$	$\frac{SS_W}{12}$	